

ATIVIDADE

SEMÁFORO

PARTICIPANTES:

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CÓDIGO FONTE:

```
// C++ code
//
int ledPedVerm = 10;
int ledPedVerd = 9;

int ledCar2Verm = 8;
int ledCar2Amar = 7;
int ledCar2Verd = 6;

int ledCar1Verm = 5;
int ledCar1Amar = 4;
int ledCar1Verd = 3;

int botaoP = 0;
int pBotao = 12;

void setup()
{
    pinMode(ledPedVerm, OUTPUT);
    pinMode(ledPedVerd, OUTPUT);

    pinMode(ledCar2Verm, OUTPUT);
    pinMode(ledCar2Amar, OUTPUT);
    pinMode(ledCar2Verd, OUTPUT);

    pinMode(ledCar1Verm, OUTPUT);
```

```

pinMode(ledCar1Amar, OUTPUT);
pinMode(ledCar1Verd, OUTPUT);

pinMode(botaoP, INPUT_PULLUP);

digitalWrite(ledCar1Verm, HIGH);
digitalWrite(ledCar2Verm, HIGH);
digitalWrite(ledPedVerm, HIGH);
}

void loop()
{
    //Ciclo Semáforo 1
    digitalWrite(ledCar1Verm, LOW);
    digitalWrite(ledCar1Verd, HIGH);
    delay(5000);
    if (digitalRead(pBotao) == HIGH){
        botaoP = 1;
    }
    digitalWrite(ledCar1Verd, LOW);
    digitalWrite(ledCar1Amar, HIGH);
    delay(2000);
    if (digitalRead(pBotao) == HIGH){
        botaoP = 1;
    }

    digitalWrite(ledCar1Amar, LOW);
    digitalWrite(ledCar1Verm, HIGH);
    if (digitalRead(pBotao) == HIGH){
        botaoP = 1;
    }
}

```

```

}
if (botaoP == 1){
    semaforoPed();
    botaoP = 0;
}

//Ciclo Semáforo 2
digitalWrite(ledCar2Verm, LOW);
digitalWrite(ledCar2Verd, HIGH);
delay(5000);
if (digitalRead(pBotao) == HIGH){
    botaoP = 1;
}

digitalWrite(ledCar2Verd, LOW);
digitalWrite(ledCar2Amar, HIGH);
delay(2000);
if (digitalRead(pBotao) == HIGH){
    botaoP = 1;
}

digitalWrite(ledCar2Amar, LOW);
digitalWrite(ledCar2Verm, HIGH);
if (digitalRead(pBotao) == HIGH){
    botaoP = 1;
}
if (botaoP == 1){
    semaforoPed();
    botaoP = 0;
}

```

```
}
```

```
void semaforoPed(){
```

```
    digitalWrite(ledPedVerm, LOW);
```

```
    digitalWrite(ledPedVerd, HIGH);
```

```
    delay(4000);
```

```
    digitalWrite(ledPedVerd, LOW);
```

```
    digitalWrite(ledPedVerm, HIGH);
```

```
    delay(500);
```

```
    digitalWrite(ledPedVerm, LOW);
```

```
    delay(500);
```

```
    digitalWrite(ledPedVerm, HIGH);
```

```
    delay(500);
```

```
    digitalWrite(ledPedVerm, LOW);
```

```
    delay(500);
```

```
    digitalWrite(ledPedVerm, HIGH);
```

```
    botaoP = 0;
```

```
}
```

